Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/06/2005
PI Name:	Law, Chung K. Ph.D.		
Project Title:	Structure and Response of Spherical Diffusion Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Chung Law's email is cklaw@princeton.edu		
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	01/01/2003	**End Date:**	01/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	Note: renewed 4 yrs/new NAG#		
Key Personnel Changes/Previous PI:	1		
COI Name (Institution):	Tse, Stephen D. (Rutgers University) Sacksteder, Kurt (NASA Glenn Research Center)		
Grant/Contract No.:	NCC3-1051		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		

Task Description:	The effort was divided into experimental and theoretical elements. The experimental effort consisted of using a porous spherical burner to issue fuel mixtures into an oxidizing environment. Different mixtures of fuel species permitted investigation of the phenomena of interest. The experiment effort was conducted primarily in the drop towers at the GRC and the NASA reduced-gravity aircraft. Space flight experiment was fully defined and approved during the program period. The theoretical effort consisted of the development of computational codes simulating the combustion phenomena of interest, and interpreting the calculated results on the basis of fundamental flame physics. The purpose of this flight definition task is to gain fundamental understanding of the structure and dynamics of diffusion flames in a simple, well-defined flow field. Using a spherical geometry this work considers: 1) gas-phase transient processes, including fuel vapor accumulation inside the flame and far-field diffusion, and their influence on flame movement and the achievement of steady flame behavior; 2) extinction boundaries of steady or near-steady flames and the identification of the associated chemistry at the global, reduced, and detailed levels; 3) the characteristics of partially-premixed flames; 4) flamefront pulsation and cellular instability especially near the extinction limits; 5) the dynamics of rotating flames; and 6) the behavior and structure of quasi-spherical flames in partial gravity environments. These studies are intended to significantly enhance the understanding of diffusion flames, especially providing capability for predicting flame behavior under varied conditions and configurations. As such, this work should contribute to engineering approaches to practical combustion systems, achieving increased energy conversion efficiencies and reductions in unwanted combustion emissions in both terrestrial and extraterrestrial environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Enhanced understanding of flame-flow vortical interaction could lead to improved burner design and operation in terms of energy efficiency and pollutant emissions.
Task Progress:	With the knowledge and experience gained from previous 1D spherical diffusion flame studies, chemically and aerodynamically more complex, 2D axisymmetric flame was investigated. The response of the rotating spherical diffusion flames with general Lewis numbers (Le) was analytically studied and microgravity experimentation was conducted at NASA Glenn to validate the theoretical findings. The rotational flow is induced by spinning the porous sphere along its axis. The flames are non-spherical because centrifugal force engenders outward flow from the burner equator, with mass continuity satisfied by inward flow to the burner poles. Such an experimental configuration assists in the studies of the effects of vortical motion in turbulent flows in flame responses, droplet spinning in the interior of combusting sprays, as well as localized flame extinction. Few analytical works have been published with no clear experimental validation. Specifically, Lozinski et al. [1] carried out an analytical work based on unity Lewis number and found strong evidence that small rotating perturbation can have significant effect on the flame dynamics as well as the extinction behavior. However since their analysis was restricted to unity Lewis number (Le), while the majority of the practical combustion systems involve hydrocarbon-based fuels that have Le that deviate from unity, it is imperative that non-unity Le analysis be carried out. An analytical model has been developed based on constant-density assumption [2]. It is shown that the flame temperature can deviate from the steady-state spherical flame temperature. Particularly, it was found that when LeF > 1 or LeO < 1, the flame temperature decreases at the poles but increases at the equator. As the Le's change to LeF < 1 or LeO > 1, the opposite happens. This is a strong indication that just by varying the Le of the mixtures either at the ambient or at the burner mixtures, the flame can extinguish either at the pole or the equator when the flame is sufficiently close to the limit condition. With the above theoretical development in mind, microgravity drop experiments were conducted at NASA Glenn to validate not only the theory but also to extend the theory by experimentally studying the flame response at high RPM, where the rotation is no longer a perturbation [3]. When subjected to rotation, it was observed that the flame deformed into a pancake shape, flattened at the poles and protruding outward from the equator, regardless of LeF and LeO. The deformation increased as the angular velocity increased but reached an asymptote at high angular velocities, at which the flame surface approached very close to the burner surface near the poles. The flame temperature, however, was noticeably influenced by the Le of the system. When the Le of the burner mixture was less than one, the emission at the poles significantly became stronger as rotation increased, even though heat loss to the burner was increasing as the flame became more flat. When this fuel mixture was moved to the ambient, just the opposite was observed. In fact, if these spherical flames were rendered to be very weak, the current results indicate that rotation could give rise to local extinction either at the poles or the equator. Finally, the flame shape deformations were shown to increase linearly with mass flow rates. All these observations were in accordance with the results of theoretical work [2], even though the theory assumes small rotational perturbations which are contrary to the large rotational velocities in the experiments. It is thus suggested that the theory does capture the essence of the effects of rotation on the flame response. References 1. Lozinski, D., and Matalon M., "Combustion of a spinning fuel droplet," Combust. Sci. Technol. 96:345-367 (1994). 2. Yoo, S. W., Qian, J., Bechtold, J. K., and Law, C. K., "Response of Spherical Diffusion Flames under Rotation with General Lewis numbers," Combustion Theory and Modelling (in press). 3. Yoo, S. W., Zhu, D. L., Law, C. K., and Sacksteder, K. R., "An Experimental Study on the Response of Spherical Diffusion Flames Subjected to Rotation and Nonequidiffusion," AIAA-2005-1141.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Yoo SW, Qian J, Bechtold JK, Law CK. "Response of spherical diffusion flames under rotation with general Lewis numbers." Combustion Theory and Modeling, In Press, February 2005. , Feb-2005

Papers from Meeting Proceedings	Yoo SW, Zhu DL, Law CK, Sacksteder KR. "An Experimental Study on the Response of Spherical Diffusion Flames Subjected to Rotation and Nonequidiffusion." 43rd AIAA Aerospace Sciences Meeting and Exhibit, Reno, Nevada, Jan. 10-13, 2005. 43rd AIAA Aerospace Sciences Meeting and Exhibit, Reno, Nevada, Jan. 10-13, 2005. Paper AIAA-2005-1141. , Jan-2005
Presentation	Yoo, S. W., Zhu, D. L., Law, C. K., and Sacksteder, K. R. "An Experimental Study on the Response of Spherical Diffusion Flames Subjected to Rotation and Nonequidiffusion" AIAA-2005-1141 Jan-2005